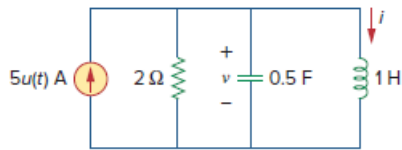


Problem

In the circuit of Fig. 16.63, find $v(t)$ and $i(t)$ for $t > 0$. Assume $v(0) = 0$ V and $i(0) = 1.25$ A.

Figure 16.63:



Step-by-step solution

Step 1 of 11

(Refer figure 16.63) the initial current across the inductor is,

$$i(0) = 1.25 \text{ A}$$

The initial voltage across the capacitor is,

$$v(0) = 0 \text{ V}$$

Step 2 of 11

Draw the Laplace equivalent of the figure 16.63 as in the Figure 1.

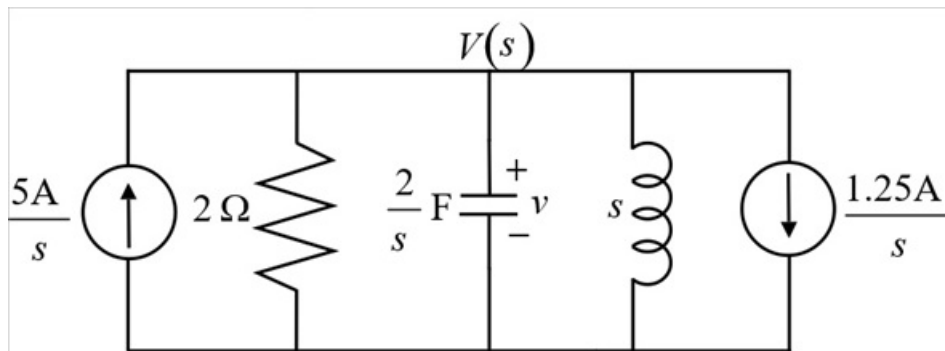


Figure 1

Step 3 of 11

Apply the nodal analysis in above figure.

$$\begin{aligned} \frac{5}{s} &= \frac{V(s)}{2} + \frac{s(V(s))}{2} + \frac{V(s)}{s} + \frac{1.25}{s} \\ \frac{3.75}{s} &= \frac{s(V(s)) + s^2(V(s)) + 2V(s)}{2s} \\ 7.5 &= s(V(s)) + s^2(V(s)) + 2V(s) \\ V(s) &= \frac{7.5}{s^2 + s + 2} \end{aligned}$$

Rewrite the above equation.

$$\begin{aligned}
 V(s) &= \frac{7.5}{s^2 + s + 2 + 0.25 - 0.25} \\
 &= \frac{7.5}{(s + 0.5)^2 + 1.75} \\
 &= \frac{7.5 \times 1.322}{\left[(s + 0.5)^2 + (1.322)^2 \right] 1.322} \\
 &= \frac{5.67 \times 1.322}{\left[(s + 0.5)^2 + (1.322)^2 \right]}
 \end{aligned}$$

The inverse Laplace transform of $\frac{a}{s^2 + a^2}$ is,

$$\frac{a}{s^2 + a^2} \leftrightarrow \sin at$$

The shifting property of Laplace transform is,

$$\frac{1}{s + a} \leftrightarrow e^{-at}$$

Then, the inverse Laplace transform of $V(s)$ is,

$$V(t) = 5.67e^{-0.5t} \sin 1.322t$$

Hence, the voltage across the capacitor v is $\boxed{5.67e^{-0.5t} \sin 1.322t \text{ V}}$.

Calculate the current $I(s)$ across the inductor.

$$I(s) = \frac{V(s)}{s} + \frac{1.25}{s} \dots\dots (1)$$

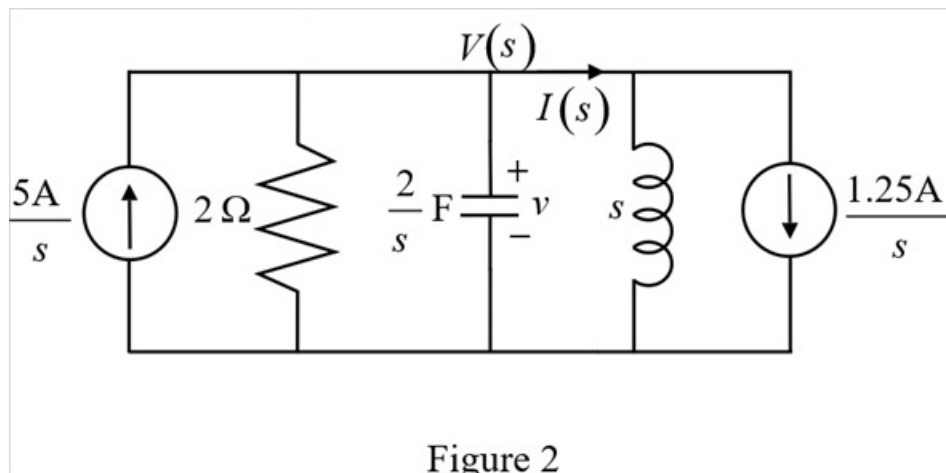


Figure 2

Substitute $\frac{7.5}{s^2 + s + 2}$ for $V(s)$ in equation (1).

$$\begin{aligned} I(s) &= \frac{7.5}{s(s^2 + s + 2)} + \frac{1.25}{s} \\ &= \frac{7.5 + 1.25(s^2 + s + 2)}{s(s^2 + s + 2)} \\ &= \frac{1.25s^2 + 1.25s + 10}{s(s^2 + s + 2)} \dots\dots (2) \end{aligned}$$

Step 8 of 11

Do the partial fraction of equation (2).

$$\frac{1.25s^2 + 1.25s + 10}{s(s^2 + s + 2)} = \frac{A}{s} + \frac{Bs + C}{(s^2 + s + 2)} \dots\dots (3)$$

$$1.25s^2 + 1.25s + 10 = A(s^2 + s + 2) + (Bs + C)s$$

$$1.25s^2 + 1.25s + 10 = s^2(A + B) + s(A + C) + 2A$$

Compare the coefficient of s^2 , s and s^0 .

$$A = 5$$

$$B = -3.75$$

$$C = -3.75$$

Step 9 of 11

Substitute 5 for A , -3.75 for B and C in equation (3).

$$\begin{aligned} \frac{1.25s^2 + 1.25s + 10}{s(s^2 + s + 2)} &= \frac{5}{s} + \frac{-3.75s - 3.75}{(s^2 + s + 2)} \\ &= \frac{5}{s} - \frac{3.75s}{(s^2 + s + 2)} - \frac{3.75}{(s^2 + s + 2)} \end{aligned}$$

Step 10 of 11

Then, rewrite the equation (2).

$$\begin{aligned} I(s) &= \frac{5}{s} - \frac{3.75s}{(s^2 + s + 2)} - \frac{3.75}{(s^2 + s + 2)} \\ &= \frac{5}{s} - \frac{3.75s}{(s + 0.5)^2 + 1.75} - \frac{3.75}{(s + 0.5)^2 + 1.75} \\ &= \frac{5}{s} - \frac{3.75s}{(s + 0.5)^2 + (1.322)^2} - \frac{3.75 \times 1.322}{[(s + 0.5)^2 + (1.322)^2]1.322} \\ &= \frac{5}{s} - \frac{3.75s}{(s + 0.5)^2 + (1.322)^2} - \frac{2.836 \times 1.322}{[(s + 0.5)^2 + (1.322)^2]} \dots\dots (4) \end{aligned}$$

Step 11 of 11

The inverse Laplace transform of $\frac{a}{s^2 + a^2}$ is,

$$\frac{a}{s^2 + a^2} \leftrightarrow \sin at$$

The inverse Laplace transform of $\frac{s}{s^2 + a^2}$ is,

$$\frac{s}{s^2 + a^2} \leftrightarrow \cos at$$

The shifting property of Laplace transform is,

$$\frac{1}{s + a} \leftrightarrow e^{-at}$$

Then the inverse Laplace transform of equation (4),

$$i(t) = \left[5 - e^{-0.5t} (3.75 \cos 1.322t - 2.836 \sin 1.322t) \right] u(t)$$

Hence, the current across the inductor is

$$\boxed{\left[5 - e^{-0.5t} (3.75 \cos 1.322t - 2.836 \sin 1.322t) \right] u(t) \text{ A} .}$$